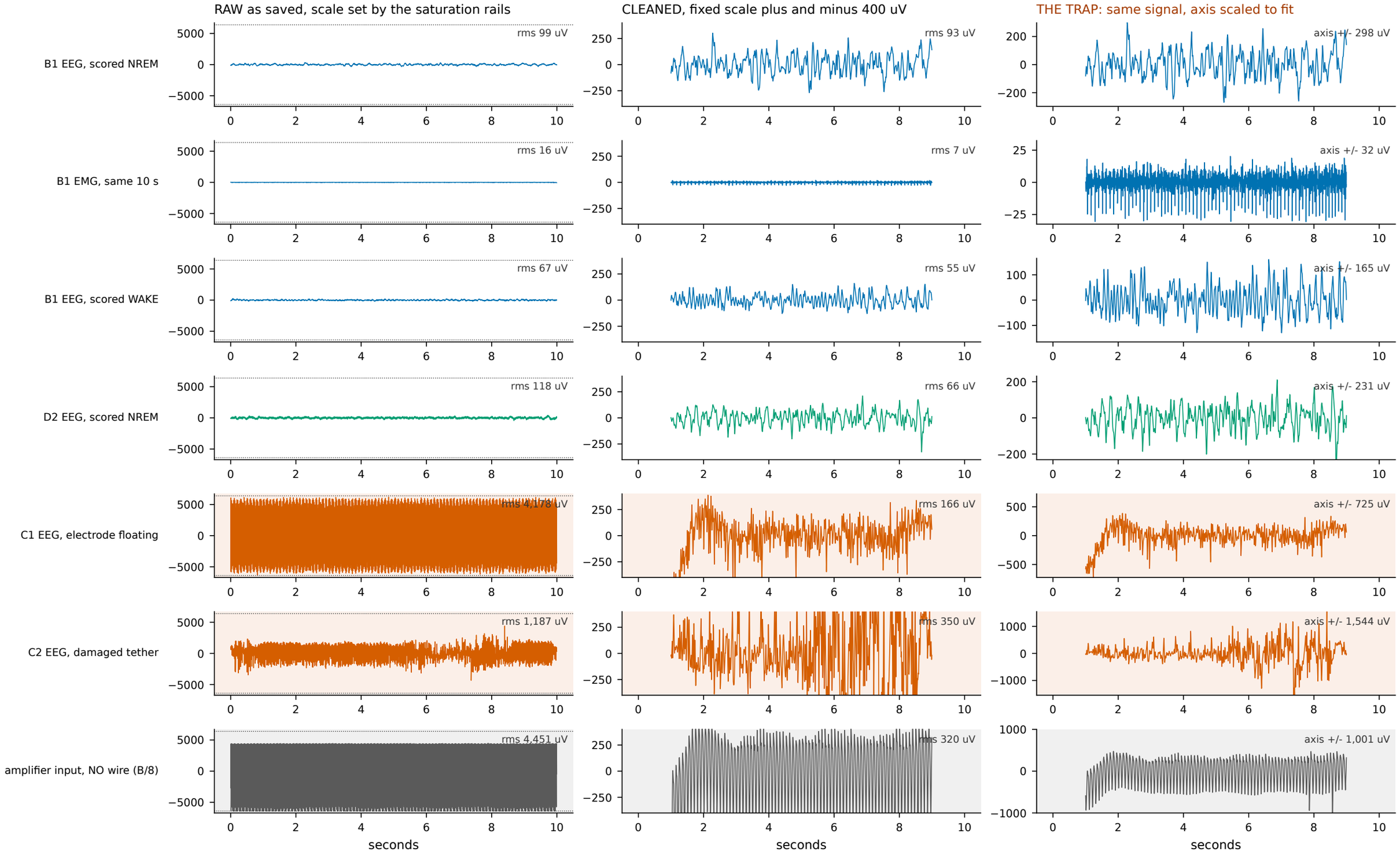
Figure 3. What the channels actually look like, one recording, one 10 second window each



PROCESSING. Left: raw as written to disk, decimated 1000 Hz to 250 Hz, microvolts, nothing else. The only filter present is the acquisition hardware band pass of 1.10 to 475.17 Hz, which cannot be removed. Middle: notch at 60, 120 and 180 Hz, quality factor 30, then a 4th order Butterworth band pass applied forwards and backwards, 0.5 to 30 Hz for EEG and 10 to 100 Hz for EMG; one second discarded at each end for the filter edge. Right: the identical cleaned signal with the axis allowed to fit, which is what plotting software does by default. WINDOW RULE, applied identically to every panel: the window whose amplitude is closest to the median of that channel, so no panel is a lucky or an unlucky moment. The floor row shows the median of twelve wire free inputs on port B, whose cleaned amplitudes span 193 to 438 microvolts; a bare wire still produces a large structured trace, and at its own scale it does not read as obviously different from data.